

AIN SHAMS UNIVERSITY
FACULTY OF ENGINEERING
Automotive Engineering Program

(MCT311s: Hydraulics and Pneumatics)
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Humanoid Arm

Pneumatic Pick-and-Place System with 3 Degrees of Freedom

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1. Introduction

Robotic arms are fundamental components in modern industrial automation, enabling repetitive, precise, and safe handling of objects in environments unsuitable or inconvenient for human operators. Pneumatic actuation, using compressed air, is a widely adopted technology in such systems due to its simplicity, high power-to-weight ratio, cleanliness, and reliability in industrial settings.

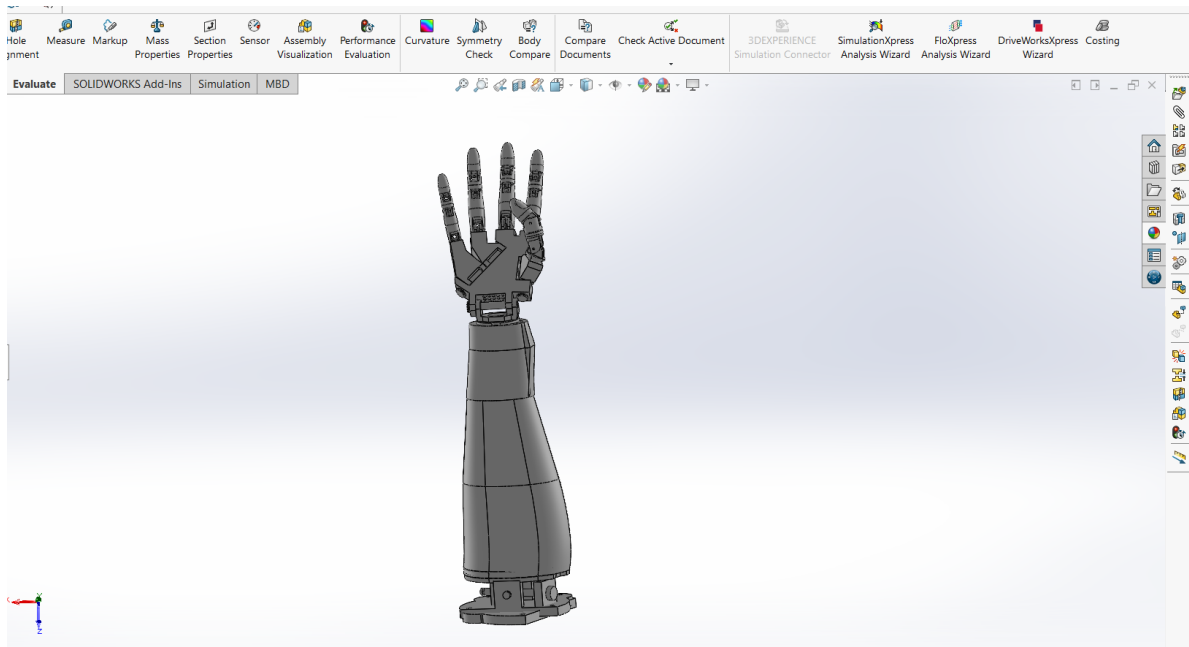
This project presents the design, simulation, and implementation of a Pneumatic Humanoid Arm capable of performing pick-and-place operations, developed as part of course MCT311s: Hydraulics and Pneumatics at Ain Shams University, Faculty of Engineering, Automotive Engineering Program. The system features three independent degrees of freedom (DOF), each actuated by a double-acting pneumatic cylinder representing the arm, wrist, and gripper functions. The system incorporates both manual and fully automatic sequential control using electrical relay logic and 30-second ON-delay timers.

The primary objectives of this project are:

- To design a functional 3-DOF pneumatic system suitable for pick-and-place tasks.
- To implement a pneumatic circuit using double-acting cylinders and 5/2 directional control valves.
- To develop an electrical control circuit using relays and 30-second ON-delay timers for both manual and automatic operation.
- To simulate the complete system in Automation Studio and validate the design before physical implementation.
- To build and demonstrate the working prototype within laboratory safety guidelines.

2. Mechanical Design:

The humanoid robotic arm is engineered with three primary Degrees of Freedom (DoF), strategically distributed across the elbow, wrist, and hand. This configuration allows for the necessary dexterity to execute precise Pick and Place operations.

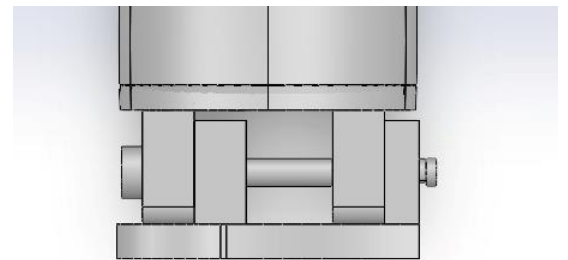


Mechanism of motion

- Elbow Joint: Wide-Range Transition

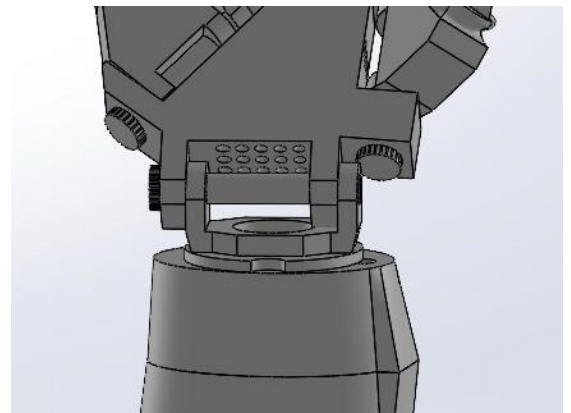
The elbow joint serves as the primary actuator for the arm's reach, facilitating the wide-range transition between the initial position and the target operational coordinates. It consists of fork joint and pin.

The used mechanism : the elbow utilizes a wire and pulley drive system. This mechanism allows for high torque transmission while keeping the distal parts of the arm lightweight.



- Wrist Joint: Pitch Orientation

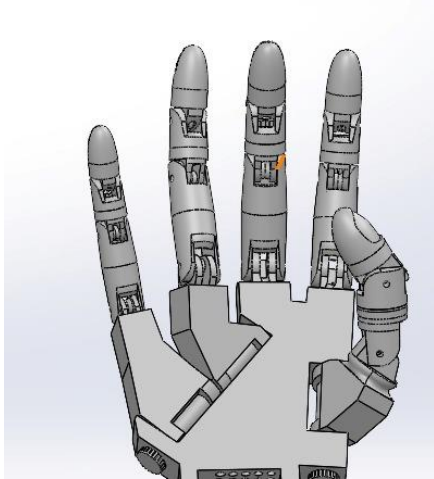
The wrist is designed as a Single-Axis Pivot Joint. It focuses on Pitch (Flexion and Extension), allowing for forward and backward pivotal movement.



- Hand Joint: End-Effector Actuation

The hand joint acts as the robot's End-Effector, responsible for the final phase of the sequence: Grasping.

It is specifically programmed to actuate the fingers to "pick" and secure the object.



Fabrication and Materials

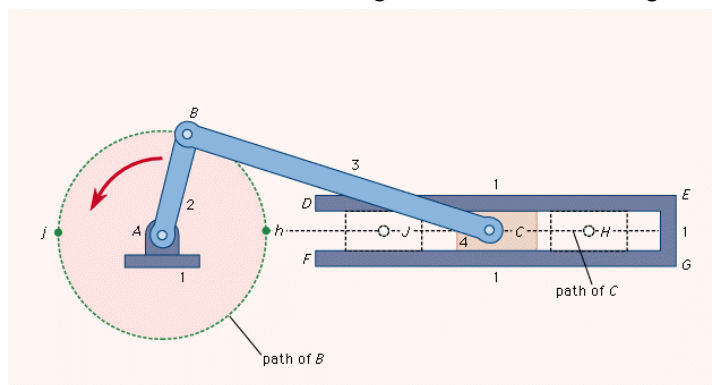
The mechanical components of the humanoid arm were fabricated using Additive Manufacturing (3D Printing) technology.

Material: All structural parts were printed using **PLA (Polylactic Acid)**.

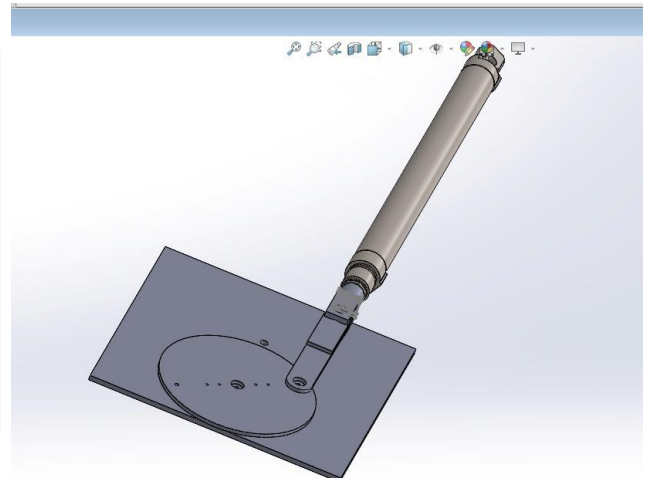
How does it work?

The question now is how does the linear motion generated by the pneumatic cylinders correlate with the arm's motion?

- 1- For hand joint we design a slider crank mechanism enabling us to extend the fingers



in one direction and retract them in another.



We designed the slide to be 10 cm long and 4 cm apart to match the cylinder stroke.

- 2- For the elbow joint, we used a 3D-printed pulley with the wire wrapped around it. The slider mechanism was laser-cut from acrylic sheets

Physical Prototype



3. Mechanical Calculations

System Pressure: 5 bar | All cylinders: Double-Acting | Stroke: 80 mm
3 cylinders total | 7 solenoid valves (5 fingers share 1 cylinder)

3.0. System Configuration

Joint / Assembly	Bore Ø	Stroke	Actuation Method
Fingers (all 5, one joint)	25 mm	80 mm	Crank + fishing-line tendon
Wrist	8 mm	80 mm	Direct pull
Elbow	32 mm	80 mm	Direct pull

3.1. Total System Load

Total arm mass: $m = 620 \text{ g} = 0.62 \text{ kg}$

Total weight:

$$W = m \times g = 0.62 \times 9.81 = \mathbf{6.08 \text{ N}}$$

Elbow – effective load component (arm at 45°):

$$W_{\perp} = W \times \sin(45^{\circ}) = 6.08 \times 0.707 = \mathbf{4.30 \text{ N}}$$

Comment: The arm is inclined at 45° at the elbow joint. Only the perpendicular component $W \times \sin(45^{\circ})$ creates a bending moment that the cylinder must resist. The axial component $W \times \cos(45^{\circ})$ is absorbed by the joint pin.

3.2. Elbow Joint – Direct Pull

Comment: The cylinder rod connects directly to a point on the forearm 100 mm from the elbow joint centre. No crank or link mechanism is used.

2.1 Torque

Attachment lever arm: $r = 0.10 \text{ m}$

$$T = W_{\perp} \times r = 4.30 \times 0.10 = \mathbf{0.430 \text{ N}\cdot\text{m}}$$

2.2 Required Force

$$F_{\text{required}} = T / r = 0.430 / 0.10 = \mathbf{4.30 \text{ N}}$$

2.3 Design Force (SF = 4)

$$F_{\text{design}} = 4.30 \times 4 = \mathbf{17.20 \text{ N}}$$

2.4 Cylinder Verification – Ø 32 mm @ 5 bar

$$A = \pi(0.032)^2 / 4 = \mathbf{8.042 \times 10^{-4} \text{ m}^2}$$

$$F_{\text{cyl}} = 500,000 \times 8.042 \times 10^{-4} = \mathbf{402 \text{ N}}$$

$$\mathbf{\text{Actual SF: } 402 / 4.30 = \approx 93.5\times}$$

Comment: The Ø 32 mm cylinder is selected from available stock. Its force greatly exceeds the static requirement, which is acceptable — the throttle valve on the exhaust controls the actual speed and force applied. The large margin also compensates for joint friction and dynamic inertia.

3.3. Wrist Joint – Direct Pull

Comment: The cylinder rod connects directly to the wrist assembly 100 mm from the wrist joint centre. No crank or link mechanism.

3.1 Load

Moving mass (wrist + hand + fingers): $m = 0.252 \text{ kg}$

$$W = 0.252 \times 9.81 = \mathbf{2.47 \text{ N}}$$

3.2 Torque

$$T = W \times r = 2.47 \times 0.10 = \mathbf{0.247 \text{ N}\cdot\text{m}}$$

3.3 Required Force

$$F_{\text{required}} = T / r = 0.247 / 0.10 = \mathbf{2.47 \text{ N}}$$

3.4 Design Force (SF = 4)

$$F_{\text{design}} = 2.47 \times 4 = \mathbf{9.88 \text{ N}}$$

3.5 Cylinder Verification – Ø 8 mm @ 5 bar

$$A = \pi(0.008)^2 / 4 = \mathbf{5.027 \times 10^{-5} \text{ m}^2}$$

$$F_{\text{cyl}} = 500,000 \times 5.027 \times 10^{-5} = \mathbf{25.15 \text{ N}}$$

$$\mathbf{\text{Actual SF: } 25.15 / 2.47 = \approx 10.2\times}$$

Comment: With direct pull, the required force is 2.47 N — half the earlier crank-based estimate of 5.49 N. The Ø 8 mm cylinder is well matched to the wrist joint.

3.4. Finger Joint – Crank + Tendon

Comment: All five fingers form a single joint driven by one Ø 25 mm cylinder. The cylinder drives a crank ($R = 0.045$ m) via a connecting link (length = 0.10 m, piston offset = 0.04 m from crank centre). The crank pulls a single fishing-line tendon that closes all five fingers simultaneously.

4.1 Load

Five fingers $\times 6$ g = 30 g = 0.030 kg
 $W_{\text{fingers}} = 0.030 \times 9.81 = \mathbf{0.294\text{ N}}$

Comment: This represents the gravitational weight of the fingers only. The primary function of this cylinder is to generate gripping force through the tendon, not just to lift the fingers.

4.2 Torque at Finger Root Joint

Effective lever arm (average finger length): $r_{\text{finger}} = 0.04$ m
 $T = W_{\text{fingers}} \times r_{\text{finger}} = 0.294 \times 0.04 = \mathbf{0.0118\text{ N}\cdot\text{m}}$

4.3 Force Required at Crank Pin

The tendon wraps around the crank at radius $R = 0.045$ m.
Using the crank radius as the moment arm (conservative, worst-case):
 $F_{\text{crank}} = T / R = 0.0118 / 0.045 = \mathbf{0.262\text{ N}}$

Comment: The crank-link geometry means the effective moment arm changes with angle. Using R directly is the worst case (minimum mechanical advantage). The safety factor of 5 covers this variation and the tendon friction.

4.4 Design Force (SF = 5)

$F_{\text{design}} = 0.262 \times 5 = \mathbf{1.31\text{ N}}$

Comment: SF = 5 is used instead of 4 to account for fishing-line friction through the finger routing channels, which cannot be precisely quantified.

4.5 Cylinder Verification – Ø 25 mm @ 5 bar

$A = \pi(0.025)^2 / 4 = \mathbf{4.909 \times 10^{-4}\text{ m}^2}$
 $F_{\text{cyl}} = 500,000 \times 4.909 \times 10^{-4} = \mathbf{245.4\text{ N}}$

Actual SF (vs crank load): $245.4 / 0.262 = \approx \mathbf{937\times}$

Comment: The cylinder force far exceeds the gravitational finger load, which is expected. The large force is used to generate grip force through the tendon, not to overcome gravity.

4.6 Estimated Grip Force

Tendon tension available: $F_{\text{tendon}} = 245.4 \text{ N}$

Force per finger (5 fingers): $245.4 / 5 = 49.1 \text{ N}$

After ~30% tendon friction loss: $49.1 \times 0.70 = \approx 34 \text{ N per fingertip}$

Comment: 34 N per fingertip is more than adequate for object manipulation. The throttle valve allows the operator to limit gripping force to avoid damaging fragile objects.

3.5. Force & Safety Factor Summary

Joint	Torque	F required	F design	F cylinder	Actual SF
Fingers – 1 cyl. (crank)	0.0118 N·m	0.262 N	1.31 N	245.4 N	~937×
Wrist – direct pull	0.247 N·m	2.47 N	9.88 N	25.15 N	~10.2×
Elbow – direct pull	0.430 N·m	4.30 N	17.20 N	402 N	~93.5×

3.6. Air Consumption

Stroke: $L = 80 \text{ mm} = 0.08 \text{ m}$ (all cylinders)

Cycle: 2 extensions + 2 retractions = 4 strokes per cycle

Absolute pressure: $P_{\text{abs}} = 5 + 1 = 6 \text{ bar}$

NL conversion: $NL = V_{\text{cycle}} [L] \times P_{\text{abs}} [\text{bar}] = V_{\text{cycle}} \times 6$

6.1 Finger Cylinder – 1 × Ø 25 mm

$V \text{ per stroke} = 4.909 \times 10^{-4} \times 0.08 = \mathbf{0.03927 \text{ L}}$

$V \text{ per cycle} = 4 \times 0.03927 = \mathbf{0.1571 \text{ L}} \rightarrow \mathbf{0.943 \text{ NL/cycle}}$

Comment: One cylinder only — all five fingers share the same joint and the same cylinder. This significantly reduces air consumption compared to one cylinder per finger.

6.2 Wrist Cylinder – 1 × Ø 8 mm

$V \text{ per stroke} = 5.027 \times 10^{-5} \times 0.08 = 4.021 \times 10^{-6} \text{ m}^3 = 0.004021 \text{ L}$

$V \text{ per cycle} = 4 \times 0.004021 = \mathbf{0.01608 \text{ L}} \rightarrow \mathbf{0.097 \text{ NL/cycle}}$

6.3 Elbow Cylinder – 1 × Ø 32 mm

$V \text{ per stroke} = 8.042 \times 10^{-4} \times 0.08 = 6.434 \times 10^{-5} \text{ m}^3 = 0.06434 \text{ L}$

$V \text{ per cycle} = 4 \times 0.06434 = \mathbf{0.2573 \text{ L}} \rightarrow \mathbf{1.544 \text{ NL/cycle}}$

6.4 Total Air Consumption

Component	Bore	Area (m ²)	Vol/stroke (L)	Vol/cycle (L)	NL/cycle
Fingers (1 × Ø 25 mm)	Ø 25 mm	4.909×10^{-4}	0.03927	0.1571	0.943
Wrist (1 × Ø 8 mm)	Ø 8 mm	5.027×10^{-5}	0.00402	0.01608	0.097
Elbow (1 × Ø 32 mm)	Ø 32 mm	8.042×10^{-4}	0.06434	0.2573	1.544
TOTAL (3 cylinders)				0.433 L	2.584 NL

Grand total: $0.943 + 0.097 + 1.544 = 2.584 \text{ NL/cycle}$

Comment: Total system uses only 3 cylinders. The elbow dominates at 60% of total air consumption due to its large bore. The fingers contribute 36% and the wrist is negligible at 4%. The faculty compressor at ~10 bar, regulated down to 5 bar, is more than sufficient for this consumption.

3.7. Pneumatic Tubing

Fingers & Wrist: Ø 6 mm OD polyurethane tubing

Elbow (5/2 valve → cylinder): Ø 8 mm OD polyurethane tubing

Comment: Larger 8 mm tubing on the elbow supply line reduces pressure drop for the high-flow Ø 32 mm cylinder. Flow velocity at Ø 8 mm ID ≈ 6 mm is approximately 4.6 m/s — well within the recommended ≤ 10 m/s limit. The 6 mm tubing is adequate for the wrist and finger cylinders.

3.8. Control Valves

Pressure regulator: One regulator set to 5 bar, supplying all joints.

Directional control: One 5/2-way solenoid valve per cylinder — 3 valves total.

Flow/speed control: One throttle valve per cylinder, meter-out configuration.

Comment: 3 cylinders = 3 solenoid valves total (fingers, wrist, elbow). Meter-out throttling (restricting exhaust) maintains back-pressure inside the cylinder during motion, giving smooth and stable actuation without jerk. This is the recommended method for pneumatic arm control.

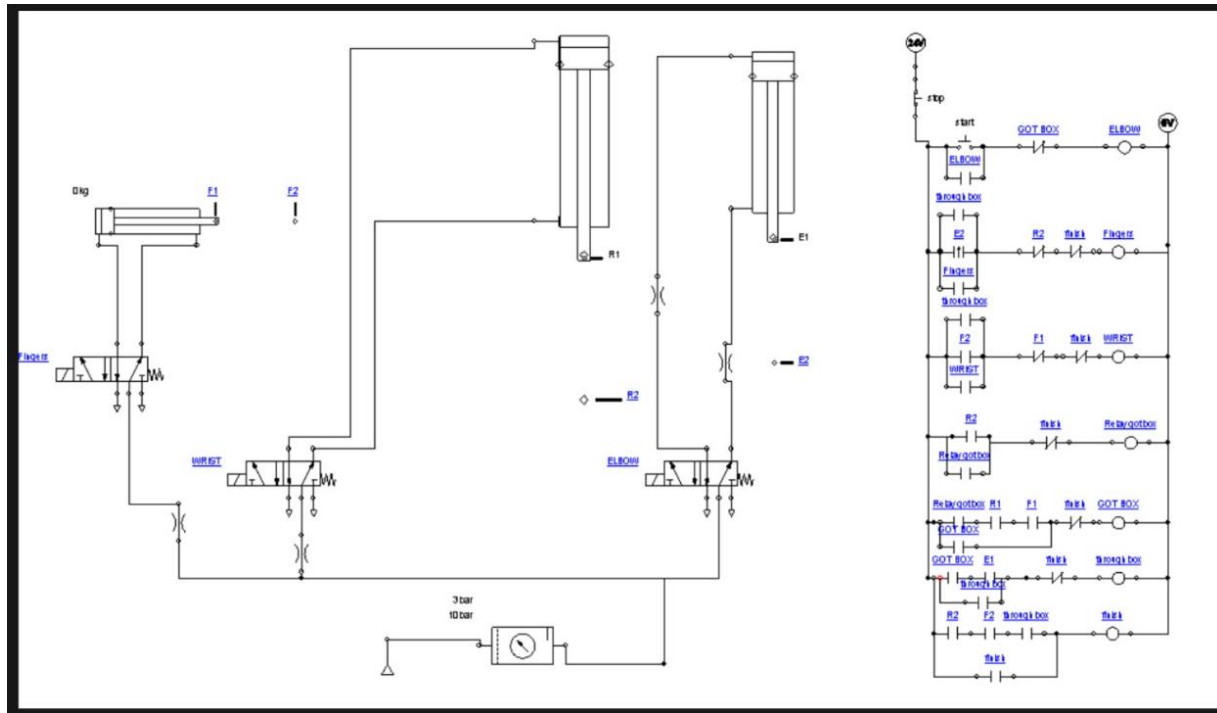
4. List of Components

Name	Function	Photo
5/2 Direction control valve	It is used to control air flow in a pneumatic system. It has five ports and two positions. It directs the airflow to control the movement of a double acting cylinder. It powered by dc voltage	
5/2 Direction control valve	It is used to control air flow in a pneumatic system. It has five ports and two positions. It directs the airflow to control the movement of a double acting cylinder. It powered by Ac voltage.	
Double acting cylinder	It acts as linear actuator	
Pneumatic Air Regulator	It is used to reduce pressure from 11 bar to circuit pressure (5 bar).	

Five-Way Union Pneumatic Connector for Tube – Model TPK	It is used to distribute the flow from the regulator into 3 paths to supply the cylinders with their flow.	
Hose	The air flows through it. We use 6 mm for the hand and wrist cylinder, and 8 mm for the elbow joint cylinder.	
Relay Module 2 Channel (5Vdc)	the relay acts as a switch that is triggered by the Arduino's output to control the extension and retraction of the cylinder.	
DC Power Adapter (24VDC - 2A)	This device powers two DCV devices because they operate on 24 volts DC.	
JSU6 Flow Control Throttle Valve	Used to control the flow rate of fluid within a system.	
Arduino Uno	Microcontroller to control the electrical circuit	

Ball Micro Limit Switch	It is used to indicate the initial and final position of the cylinder stroke.	
Omron Press Button Cap12x12	We use 2 presses for start and stop the system.	
Breadboard and jumpers	Used for electrical connection	

5. Electro_Pneumatic circuit



Control Logic and Operational Sequence

The system follows a closed-loop sequential control logic, where **Limit Switches** are utilized at the end of each cylinder's stroke to trigger the next phase of motion. This ensures high reliability and precise coordination between the three Degrees of Freedom (DoF).

Sequence of Operation

The "Pick and Place" cycle is executed through the following automated steps:

1. **Reaching Phase (Elbow):**

The cycle begins with the **Elbow Cylinder** extending.

Once it reaches the end of its stroke, it triggers a **Limit Switch E1**.

2. **Grasping/Placement Phase (Hand):**

The Elbow's limit switch signal actuates the **Hand Solenoid (Sol2)**.

The Hand Cylinder extends to perform the "Pick" or "Place" action.

Upon reaching the full stroke, the Hand's **Limit Switch F1** is triggered.

3. **Orientation Phase (Wrist):**

The signal from the Hand's limit switch actuates the **Wrist Solenoid (Sol3)**.

The Wrist Cylinder extends (Pitch movement) to finalize the object placement. After each phase of motion, there is a delay 5 seconds between each DOF to ensure stability.

4. **Retraction Phase (Resetting):**

After the delay, the **Wrist Cylinder** retracts.

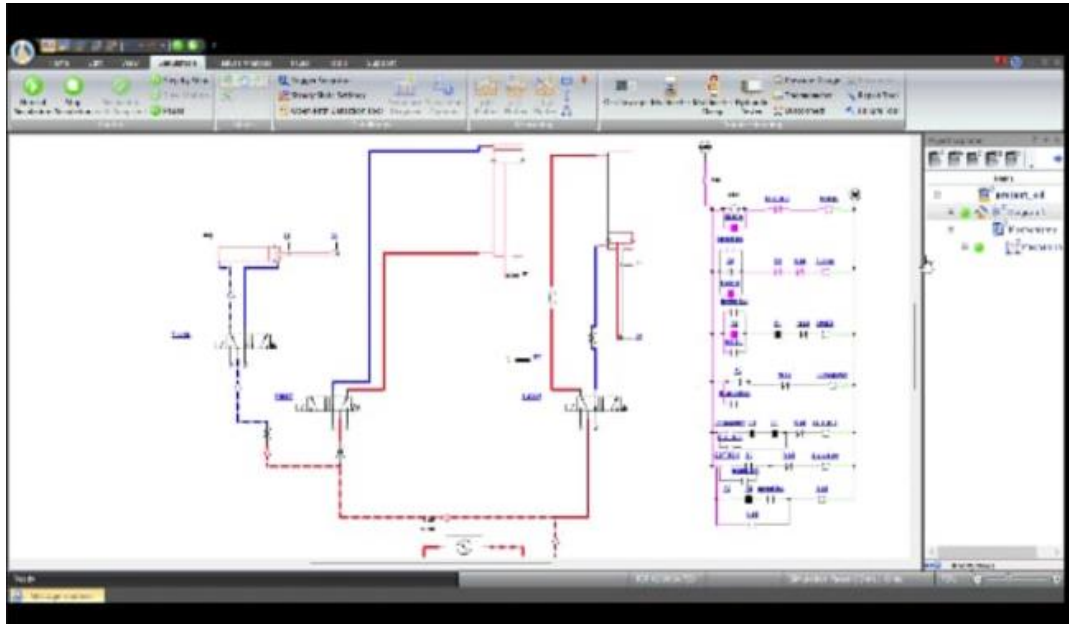
Once it hits the **Limit Switch at the start of its stroke**, it signals the **Hand Cylinder** to retract.

When the Hand reaches its initial position limit switch, the **Elbow Cylinder** finally retracts.

5. Cycle Completion:

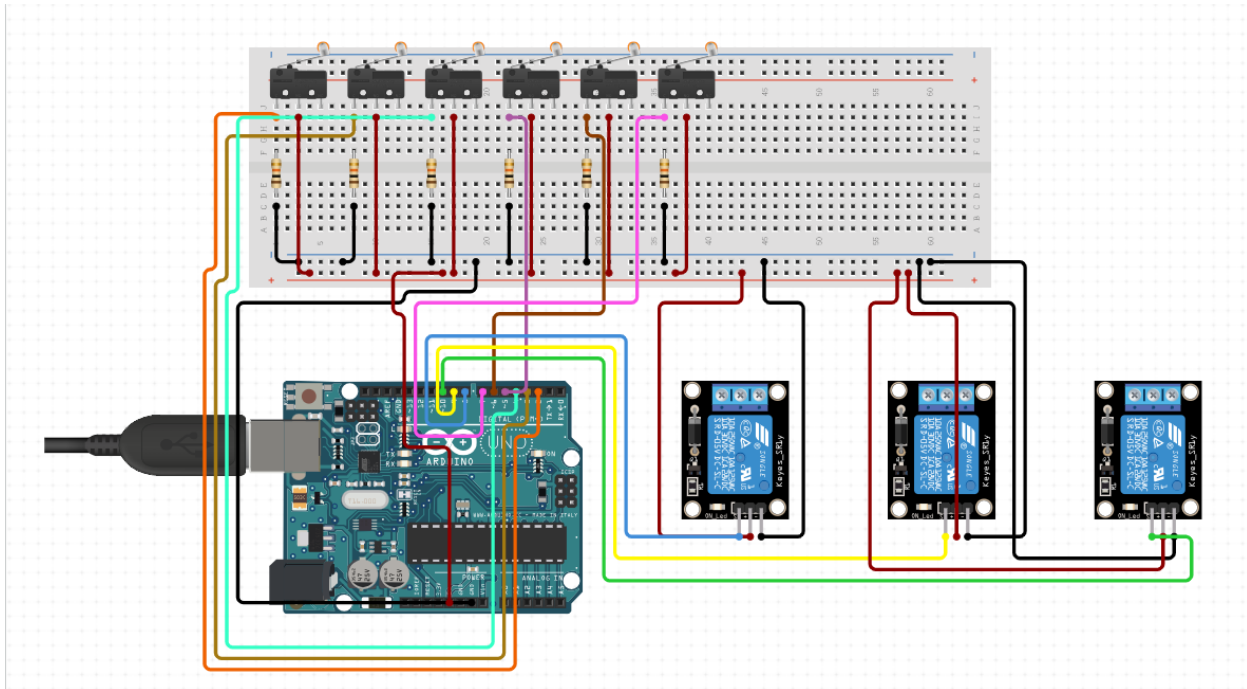
The arm returns to its **Initial Position**. The entire sequence repeats automatically until the **Stop Switch** is pressed by the operator.

Simulation



6. Electrical Circuit

The electrical system acts as the "nervous system" of the humanoid arm, interfacing the microcontroller with the pneumatic actuators and feedback sensors. The circuit is designed to handle signal processing and power distribution safely.



Control Unit: Arduino Uno

An **Arduino Uno** is used as the central control unit. It receives input signals from the sensors and executes the sequential logic to drive the outputs (solenoid valves).

Feedback System: Limit Switches

To achieve precise sequential control, the circuit integrates **6 Limit Switches**: Two switches are dedicated to each cylinder: one to detect the **Start of Stroke** and another for the **End of Stroke**.

Actuation Interface: Relay Modules

Relays act as electrically operated switches that allow the low-power Arduino signal to control the high-power solenoid valves.

7. Arduino Code

```
#define START_BTN 13
#define STOP_BTN 12
#define R1 8
#define R2 9
#define R3 10
#define C1_EXT 2
#define C1_RET 7
#define C2_EXT 6
#define C2_RET 5
#define C3_EXT 4
#define C3_RET 3

#define STEP_DELAY 5000

bool systemRunning = false;
```

```
// ===== SETUP =====  
  
void setup() {  
  pinMode(START_BTN, INPUT_PULLUP);  
  pinMode(STOP_BTN, INPUT_PULLUP);  
  pinMode(R1, OUTPUT);  
  pinMode(R2, OUTPUT);  
  pinMode(R3, OUTPUT);  
  pinMode(C1_EXT, INPUT_PULLUP);  
  pinMode(C1_RET, INPUT_PULLUP);  
  pinMode(C2_EXT, INPUT_PULLUP);  
  pinMode(C2_RET, INPUT_PULLUP);  
  pinMode(C3_EXT, INPUT_PULLUP);  
  pinMode(C3_RET, INPUT_PULLUP);  
  Serial.begin(9600);  
  stopAll();  
}
```

```
// ===== LOOP =====

void loop() {
    if (digitalRead(START_BTN) == LOW && !systemRunning) {
        systemRunning = true;
        Serial.println("System STARTED");
        delay(300);
    }
    if (digitalRead(STOP_BTN) == LOW && systemRunning) {
        systemRunning = false;
        stopAll();
        Serial.println("System STOPPED");
        delay(300);
    }
    if (systemRunning) {
        runSequence();
    }
}

-
// ===== FUNCTIONS =====

void stopAll() {
    digitalWrite(R1, HIGH);
    digitalWrite(R2, HIGH);
    digitalWrite(R3, HIGH);
}

bool checkStop() {
    if (digitalRead(STOP_BTN) == LOW) {
        delay(50);
        if (digitalRead(STOP_BTN) == LOW) {
            systemRunning = false;
            stopAll();
            Serial.println("EMERGENCY STOP");
            delay(300);
            return true;
        }
    }
    return false;
}

```

```

bool safeDelay(int ms) {
    for (int i = 0; i < ms; i += 50) {
        delay(50);
        if (checkStop()) return false;
    }
    return true;
}

bool waitFor(int pin) {
    while (digitalRead(pin) == HIGH) {
        if (checkStop()) return false;
    }
    return true;
}

// ===== SEQUENCE =====

void runSequence() {

    // --- C1 EXTEND ---
    Serial.println("C1 EXTEND");
    digitalWrite(R1, LOW);
    if (!waitFor(C1_EXT)) return;
    if (!safeDelay(STEP_DELAY)) return;

    // --- C2 EXTEND ---
    Serial.println("C2 EXTEND");
    digitalWrite(R2, LOW);
    if (!waitFor(C2_EXT)) return;
    if (!safeDelay(STEP_DELAY)) return;

    // --- C3 EXTEND ---
    Serial.println("C3 EXTEND");
    digitalWrite(R3, LOW);
    if (!waitFor(C3_EXT)) return;
    if (!safeDelay(STEP_DELAY)) return;
}

```

```
// --- C2 RETRACT ---  
Serial.println("C2 RETRACT");  
digitalWrite(R2, HIGH);  
if (!waitFor(C2_RET)) return;  
if (!safeDelay(STEP_DELAY)) return;  
  
// --- C3 RETRACT ---  
Serial.println("C3 RETRACT");  
digitalWrite(R3, HIGH);  
if (!waitFor(C3_RET)) return;  
if (!safeDelay(STEP_DELAY)) return;  
  
// --- C1 RETRACT ---  
Serial.println("C1 RETRACT");  
digitalWrite(R1, HIGH);  
if (!waitFor(C1_RET)) return;  
if (!safeDelay(STEP_DELAY)) return;
```

```

// --- C3 EXTEND AGAIN ---
Serial.println("C3 EXTEND (2nd)");
digitalWrite(R3, LOW);
if (!waitFor(C3_EXT)) return;
if (!safeDelay(STEP_DELAY)) return;

// --- C2 EXTEND AGAIN ---
Serial.println("C2 EXTEND (2nd)");
digitalWrite(R2, LOW);
if (!waitFor(C2_EXT)) return;
if (!safeDelay(STEP_DELAY)) return;

// --- C2 RETRACT AGAIN ---
Serial.println("C2 RETRACT (2nd)");
digitalWrite(R2, HIGH);
if (!waitFor(C2_RET)) return;
if (!safeDelay(STEP_DELAY)) return;

// --- C3 RETRACT AGAIN ---
Serial.println("C3 RETRACT (2nd)");
digitalWrite(R3, HIGH);
if (!waitFor(C3_RET)) return;
if (!safeDelay(STEP_DELAY)) return;

Serial.println("Cycle COMPLETE — restarting...");
}

```

8. Conclusion

In conclusion, this project has been a successful journey in bridging the gap between theoretical mechanical principles and practical engineering applications. By designing and building this pneumatic robotic arm, we were able to apply the core concepts of fluid power, mechanical design, and automated control that we studied throughout the course.

The project evolved from an initial conceptual model into a fully functional pneumatic system. Utilizing Arduino as the central control unit proved to be a highly effective and cost-efficient solution. Its ability to interface seamlessly with solenoid valves allowed for precise actuation and

synchronized movement of the cylinders, demonstrating how microcontrollers can bring intelligence to traditional fluid power systems.

Beyond the technical achievements—such as calculating air consumption and designing the slider mechanism—this experience was invaluable for our professional growth.

9. Contribution Sheet

Student Name	Student ID	Contribution
Taha Gamal Alieldin	2100566	%16
Fatma Samy Ahmed	2100660	15%
Amr Yasser	2100486	15%
Mohamed Tag Ibrahim	2101756	14%
Ahmed Ismail	2100722	14%
Kholoud Ahmed	2100712	13%
Mohamed Khaled	2200721	13%

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11. Link of Simulation:

https://drive.google.com/drive/folders/1EAILHo0TKiAo2lXQEAJaouZRjdBGVw_0?usp=sharing
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